LEARNER GUIDE

When two lines meet

Straight Lines and Connected Representations. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Last lesson you saw that parallel lines never meet. This lesson is about what happens when two lines have different gradients: they cross at exactly one point. In a real story about money, that crossing is the break-even point — the moment two options cost exactly the same. You will read break-even points off graphs, slide a marker until the gap between two plans closes to zero, work out which option is cheaper on each side of the crossing, and use all of it to decide between two real deals. The question you are exploring is: what does the point where two lines cross tell us?

What you are learning to notice

- That two lines with different gradients cross at exactly one point.
- That at the crossing the two y-values are equal — both options cost the same.
- That the x of the crossing tells you HOW MANY, and the y tells you the COST there.
- That the cheaper option swaps over at the crossing: one wins before it, the other after.

Moving through it, one step at a time

In Cost Comparison, read a break-even point from two cost lines — its x (how many units) and its y (the cost). In Slide to the crossing, drag the marker and watch the gap between the two plans; when the gap is zero you have found the break-even. In What does the crossing mean?, decide which plan is cheaper for a few units and for many. Then use Which deal wins? to find a break-even and advise someone who will use a little or a lot. There is no rush; use the coordinate grid on the Scratchpad to draw two lines and mark where they cross.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (working out a cost from an equation (substituting a value of x), reading coordinates, and knowing that at a crossing the two lines share the same point), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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